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28.

Orbital	n	l	Radial node $n - l - 1$	Angular node l
2s	2	0	1	0
4p	4	1	2	1
5d	5	2	2	2
4f	4	3	0	3

30. i) ground state

1	1	1	1	1

ii) maximum exchange energy

1	1	1	1	1

32.

Orbital	n	l
$3p_x$	3	1
$4d_{x^2-y^2}$	4	2

34.

$$\Delta x.\,\Delta p \geq \frac{h}{4\pi}$$

$$\Delta x.\,\Delta p \geq 5.28\times10^{-35}\,Kgm^2s^{-1}$$

$$\Delta x.\,(m\Delta v) \geq 5.28\times10^{-35}\,Kgm^2s^{-1}$$

Given $\Delta v = 0.1\%$

$v = 2.2 \times 10^6\ ms^{-1}$

$m = 9.1 \times 10^{-31}\ Kg$

$$\Delta v = \frac{0.1}{100}\times2.2\times10^6\,ms^{-1}$$

$$= 2.2\times10^3\,ms^{-1}$$

$$\therefore \Delta x \geq \frac{5.28\times10^{-35}\,Kgm^2s^{-1}}{9.1\times10^{-31}\,Kg\times2.2\times10^3\,ms^{-1}}$$

$$\Delta x \geq 2.64\times10^{-8}\,m$$

35. Electronic configuration of oxygen

$= 1s^2\ \ 2s^2\ \ 2p^4$

⇅ ⇅ ⇅ 1 1

$\therefore$ 8th electron present in $2p_x$ orbital and the quantum numbers are

$n = 2, l = 1, m_l =$ either +1 or -1 and $s = -1/2$

Electronic configuration of chlorine =

$1s^2$	$2s^2$	$2p^6$			$3s^2$	$3p^5$		
⇅	⇅	⇅	⇅	⇅	⇅	⇅	⇅	1
						$3p_x$	$3p_y$	$3p_z$

15th electron present in $3P_Z$ orbital and the quantum numbers are n = 3, l = 1, m_l =either +1 or -1 and m_s = +1/2

36.

$$E_n = \frac{-13.6}{n^2}\,eV\ \ atom^{-1}$$

$$n = 3\quad E_3 = \frac{-13.6}{3^2} = \frac{-13.6}{9} = -1.51\ eV\ atom^{-1}$$

$$n = 4\quad E_4 = \frac{-13.6}{4^2} = \frac{-13.6}{16} = -0.85\ eV\ atom^{-1}$$

$\Delta E = (E_4 - E_3) = (-0.85) - (-1.51)$ eV atom^{-1}

$= (-0.85 + 1.51)$

$= 0.66$ eV atom^{-1}

$(1eV = 1.6 \times 10^{-19}\ J)$

$\Delta E = 0.66 \times 1.6 \times 10^{-19}\ J$

$\Delta E = 1.06 \times 10^{-19}\ J$

$hv = 1.06 \times 10^{-19}\ J$

$$\frac{hc}{\lambda} = 1.06\times10^{-19}\,J$$

$$\therefore \lambda = \frac{hc}{1.06\times10^{-19}\,J}$$

$$= \frac{6.626\times10^{-34}\,JS\times3\times10^8\,ms^{-1}}{1.06\times10^{-19}\,J}$$

$$\lambda = 1.875\times10^{-6}\,m$$